Roadmap and Team Plan

What is done, the plan for the remaining hours, and what comes next

CoolRoute: Pedestrian Routing with Shade and Heat-Exposure Budgets

1. What is already done

Item	Status
Routing engine (shortest and heat-budget)	Done and tested
Browser prototype with all scenario controls	Done
JSON backend with errors and warnings	Done and tested
Charts, formulas, ML demo	Done
Ten PDFs	Done

2. Plan for the remaining hackathon hours

When	Task	Owner	Output
Hours 0 to 4	Run the prototype, read files 01, 06, 07. Agree the story and who speaks	Whole team	Shared understanding
Hours 4 to 10	Rehearse the demo and the 5-minute script; fix any wording	Presenter	Smooth demo
Hours 10 to 18	Optional upgrade: load a real few-block area of Jaipur with OSMnx and show it as a second map	Developer	Real-data screenshot
Hours 18 to 24	Optional upgrade: fetch live temperature from Open-Meteo and scale the sun factor	Developer	Weather-aware heat
Hours 24 to 30	Practise Q&A; with the file 09 questions; time the pitch	Whole team	Confident answers
Hours 30 to 36	Freeze the build. Backup screenshots. Print files. Sleep	Whole team	Ready to present

3. Roadmap after the hackathon

Step	What	Success looks like
1	Real Jaipur street graph from OpenStreetMap	Routes on real streets
2	Tree and building shade from open data, trained shade model	Shade error measured against field checks
3	Live weather and sun position (pvlib)	Heat score follows the actual day
4	Walk test with students on campus	People prefer CoolRoute; model matches how it felt
5	Exact budget method and contraction hierarchies	Optimal routes, fast on a whole city
6	Audio and large-text accessibility	Usable by people with low vision
7	Pilot with a city office	A paid licence

4. Files in the project folder

```
coolroute.html        interactive prototype (open in any browser)
core.py / core.js     routing engine in Python and JavaScript (same maths)
api_server.py         JSON backend
test_core.py          13 engine tests
test_api.py           12 backend tests
verify.js             JavaScript versus Python check
make_figs.py          charts and experiments
pdfs/                 these ten PDF files
```

5. Setup in one minute

- **Prototype:** open `coolroute.html`.
- **Engine tests:** `python3 test_core.py`.
- **Backend:** `python3 -m uvicorn api_server:app --port 8765`, then call /api/v1/health.
- Python needs numpy, matplotlib, scikit-learn and reportlab only for the charts and PDFs, not for the engine or backend.

6. Risks for the next hours

Risk	Plan
Venue internet is slow	The prototype runs with no internet. Fonts fall back to system fonts
A judge asks for real-city results	Say plainly the prototype is synthetic; show the real-data plan; if time allows, add the OSMnx map
Demo laptop fails	Open the prototype link on a phone; keep screenshots
Time runs short in the pitch	Skip the maths slide; keep demo, proof and one-line business model